

Chapter 18 - Equilibrium

Key Concepts

● 18.1 Rates of Reaction

- The rate of a chemical reaction is usually expressed as the change in the amount of a reactant per unit time.
- Factors that influence the rate of a chemical reaction are temperature, concentration, particle size, and the use of a catalyst.

● 18.2 Reversible Reactions and Equilibrium

- Chemical equilibrium is a state in which the forward and reverse reactions take place at the same rate. At chemical equilibrium, no net change occurs in the system.
- Stresses that can alter the equilibrium position of a reaction are changes in the concentrations of reactants or products, changes in temperature, and changes in pressure.
- A K_{eq} value greater than 1 means that products are favored over reactants; a K_{eq} less than 1 means that reactants are favored over products.

● 18.3 Solubility Equilibrium

- The smaller the numerical value of the solubility product constant, the lower the solubility of the compound.
- A precipitate will form if the product of the concentrations of two ions in a mixture is greater than the K_{sp} of the compound formed from the ions.

● 18.4 Entropy and Free Energy

- Spontaneous reactions produce substantial amounts of product and release free energy.
- An increase in entropy favors the spontaneous reaction; a decrease favors the nonspontaneous reaction.
- The size and direction of enthalpy changes and entropy changes together determine whether a reaction is spontaneous.
- The numerical value of ΔG is negative in spontaneous processes because the system loses free energy.

● 18.5 The Progress of Chemical Reactions

- The value of the specific rate constant, k , in the rate equation is large if products form quickly from reactants.
- Graphs of the progress of a reaction show peaks that correspond to the energies of activated complexes and valleys that correspond to the energies of intermediates and products.

Vocabulary

- activated complex (p. 544)
- activation energy (p. 543)
- chemical equilibrium (p. 550)
- collision theory (p. 542)
- common ion (p. 563)
- common ion effect (p. 563)
- elementary reaction (p. 578)
- entropy (p. 569)
- equilibrium constant (p. 556)
- equilibrium position (p. 551)
- first-order reaction (p. 576)
- free energy (p. 566)
- Gibbs free-energy change (p. 572)
- inhibitor (p. 547)
- intermediate (p. 578)
- law of disorder (p. 569)
- Le Châtelier's principle (p. 552)
- nonspontaneous reaction (p. 567)
- rate (p. 542)
- rate law (p. 575)
- reaction mechanism (p. 578)
- reversible reaction (p. 549)
- solubility product constant (p. 561)
- specific rate constant (p. 575)
- spontaneous reaction (p. 567)
- transition state (p. 544)

Key Equations

$$K_{eq} = \frac{[C]^c \times [D]^d}{[A]^a \times [B]^b}$$

$$\Delta G = \Delta H - T\Delta S$$

$$\text{rate} = k[A]^a[B]^b$$